

Problem 2.21

Show that if $\delta_t = \delta$ for all t then

$$i_n = \frac{a(n) - a(n-1)}{a(n-1)} = e^\delta - 1.$$

Letting

$$i = e^\delta - 1,$$

show that

$$a(t) = (1 + i)^t.$$

Solution

Since

$$i_n = \frac{a(n) - a(n-1)}{a(n-1)},$$

and letting

$$i = e^\delta - 1,$$

we have

$$\frac{a(n) - a(n-1)}{a(n-1)} = i.$$

Therefore,

$$\frac{a(n)}{a(n-1)} - 1 = i,$$

so

$$\frac{a(n)}{a(n-1)} = i + 1.$$

Hence,

$$a(n) = (i + 1)a(n-1).$$

We see that this depends on the previous value, so similarly,

$$a(n-1) = (i + 1)a(n-2),$$

and

$$a(n-2) = (i + 1)a(n-3).$$

Substituting repeatedly,

$$a(n) = (i + 1)^2 a(n-2) = (i + 1)^3 a(n-3) = \cdots = (i + 1)^n a(0).$$

Since an accumulation function satisfies $a(0) = 1$, it follows that

$$a(n) = (i + 1)^n a(0) = (i + 1)^n.$$

Therefore,

$$a(n) = (1 + i)^n.$$

Hence, writing the same form for time t ,

$$a(t) = (1 + i)^t.$$